

BASE EXPANSION WORKS AND ENVIRONMENTAL MANAGEMENT

Work on Dampier

Base expansion commenced on 9 October and will be largely complete by June 2001, involving a capital budget of \$13m.

The principle activities and facility developments involved in the expansion are:

A. DREDGING

Approximately 700,000 m³ of material is to be dredged in King Bay to form an entrance channel, vessel berths, cyclone moorings and to provide access to the slipway.

The experience of Woodside constructing their nearby base in 1981 indicates that two types of dredges will be required, a Cutter Suction to remove the soft unconsolidated material (approx.70%) and a Dipper Dredge (barge mounted back-hoe) to remove harder consolidated material.

The Cutter Suction dredge size will be deliberately modest due to onshore spoil management requirement and environmental considerations.

The Dipper Dredge will be the largest of its type in the world, and will be an ideal remedial dredging tool using the experience gained from the earlier Woodside project.

The layout of the Base has been very much driven by the desire to avoid or minimize blasting while fulfilling functional objectives.

The entrance channel into the Mermaid Base will be 30 m wide and dredged to 6 m below chart datum. The dredge spoil will be pumped ashore and used as fill around the Base.

Dredges are expected to be onsite for approximately 7 months commencing mid November.

B. QUAY WALL (BERTH 1)

Market research and customer needs have caused Mermaid to relocate and redesign the main berth to accommodate a wider range of vessels than originally contemplated. The berth is now located in deeper water with better vessel access.

The regional offshore fleet characteristics have been changing in terms of vessel size. There are now four vessels operating in the region with 12,000 to 18,000 hp. When design commenced there were none of this size.

The depth alongside Berth 1 will be 7.5m. King Bay has a statistical average extreme low tide (MLWS) of 0.9 m, the occurrence of which can be expressed in hours per month. The largest